



Master Specification: Driver Module (Mobile App)

Role Tier: Level 3 (Field Execution)

Reporting To: Dispatcher

Primary Objective: Safe, efficient transport of goods and capturing "Proof of Delivery" (PoD).

1. The Interactive Manifest (Daily Agenda)

The driver starts their day by "Binding" themselves to a vehicle and a route.

- **Vehicle Check-in:** A digital checklist (Fuel level, Tire pressure, Odometer) that must be completed before the route starts. This data goes back to the **Logistics Manager**.
- **Sequence List:** A vertical list of all stops, optimized by the **Dispatcher**.
 - Each stop shows: Customer Name, Address, Package Count, and "Special Instructions" (e.g., "Beware of dog" or "Leave at back gate").
- **Load Verification:** The driver scans each package as it's loaded from the **Warehouse Manager's** dock to ensure nothing is missing before they leave the geofence.

2. Navigation & Geofence Integration

- **Turn-by-Turn GPS:** Integrated Google Maps navigation that follows the **Dispatcher's** optimized path.
- **Auto-Arrive:** When the driver's phone enters the 50m geofence of a customer's address, the app automatically:
 1. Pops up the delivery details.
 2. Sends a "Driver is outside" notification via the **Customer Support Bot**.
- **Deviation Logging:** If the driver goes off-route for more than 10 minutes, the app prompts for a reason (e.g., "Roadblock," "Fueling") which is sent to the **Dispatcher**.

3. Proof of Delivery (PoD) & Completion

This is the most important data point for the **Logistics Manager's** financial records.

- **Multi-Modal Verification:**
 - **Photo Capture:** A photo of the package at the doorstep.
 - **E-Signature:** Digital signature on the glass.
 - **QR Scan:** Scanning the package one last time to "Close" the order.
- **COD (Cash on Delivery) Handling:** If the order is "Pay on Delivery," the app opens a secure portal to log the payment or scan a customer's payment QR code.

4. Driver Wallet & Performance Score

- **Live Earnings:** A "Wallet" view showing how much they've earned today (Base pay + Tips + Delivery bonuses).
- **Personal Scorecard:** Shows their current rating (from the **Customer Support** feedback) and their "On-Time" percentage.
- **Support Chat:** A quick-access button to talk to the **Dispatcher** or the **AI Bot** if they can't find a location.

5. Generative AI Integration (The "Driver's Assistant")

How GenAI helps the driver stay hands-free and safe:

- **Voice-Activated Commands:** The driver can say, *"Hey AI, read me the delivery instructions for the next stop,"* or *"Report a flat tire to the manager."*
- **Smart Parking Assistant:** The AI analyzes historical data from other drivers and suggests: *"Park on the side street; the main entrance is usually blocked by construction."*
- **Automated Log Summaries:** At the end of the shift, the AI drafts the "End of Day" report for the driver: *"You completed 22 deliveries, saved 10% on fuel by following the route, and earned a 5-star average today."*

Technical Hand-shake for your Team:

From Driver	To System	Data Transferred
GPS Pings	Dispatcher / Manager	Real-time coordinates every 30 seconds.
PoD Image	Customer Support / WM	High-res JPEG + Timestamp + Lat/Long.
Fuel/Expense Upload	Logistics Manager	OCR (Optical Character Recognition) of receipts.

Final System Overview (All 6 Roles)

1. **Logistics Manager:** Sets up the system, geofences, and roles.
2. **Vendor:** Sends goods to the warehouse.
3. **Warehouse Manager:** Maps the bins, receives goods, and packs orders.
4. **Customer Support (Bot):** Takes orders and answers "Where is my stuff?"

5. **Dispatcher:** Groups orders and assigns the best route to the driver.
6. **Driver:** Follows the GPS, delivers the package, and takes a photo.